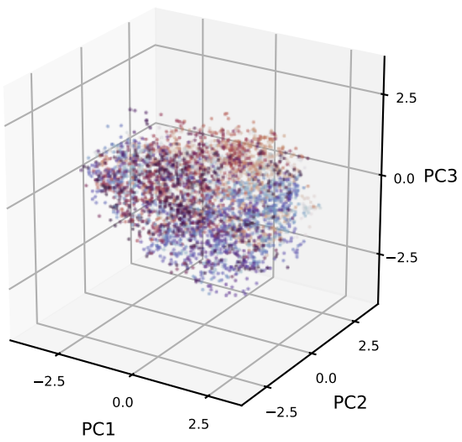
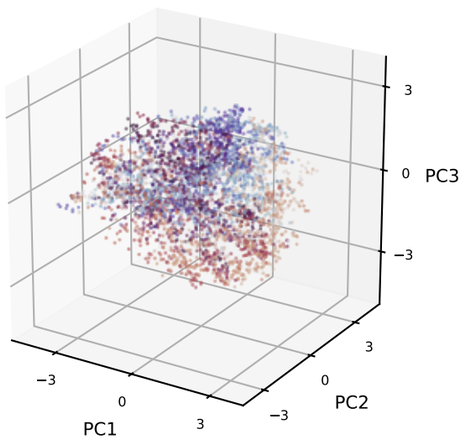
All eight seed pairs · frozen LSTMs at 1B frames · PCs 1–3
3,393 matched states: varying movement, facing, and remembered goals

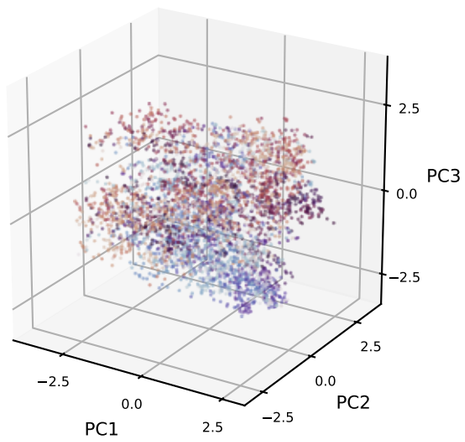
Seed 1 · Grid only
PC1–3: 24.9% of variance



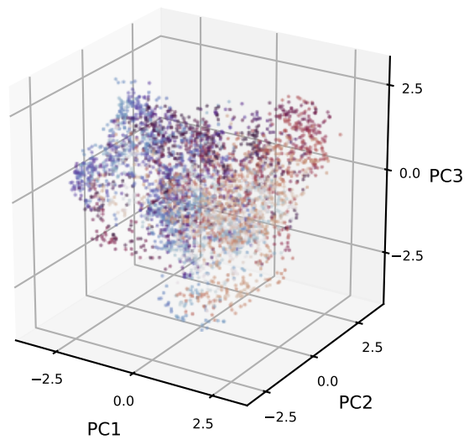
Seed 1 · PGC + grid
PC1–3: 26.5% of variance



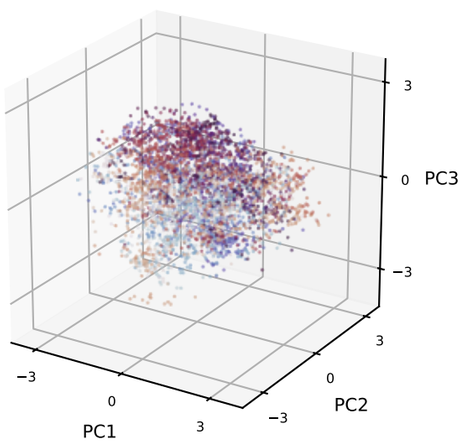
Seed 2 · Grid only
PC1–3: 24.3% of variance



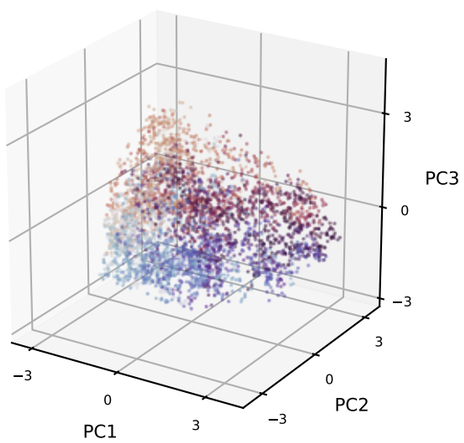
Seed 2 · PGC + grid
PC1–3: 27.1% of variance



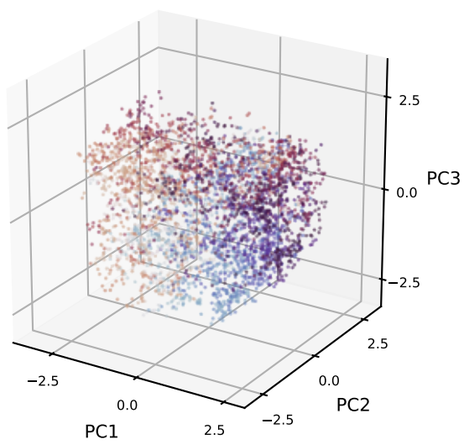
Seed 3 · Grid only
PC1–3: 22.6% of variance



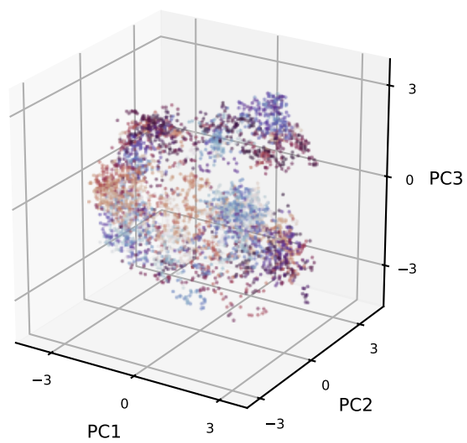
Seed 3 · PGC + grid
PC1–3: 29.5% of variance

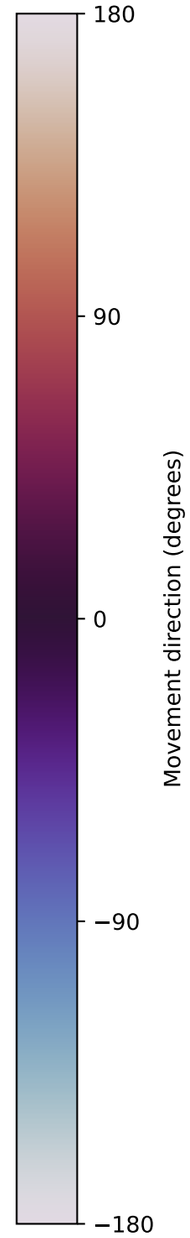


Seed 4 · Grid only
PC1–3: 25.2% of variance

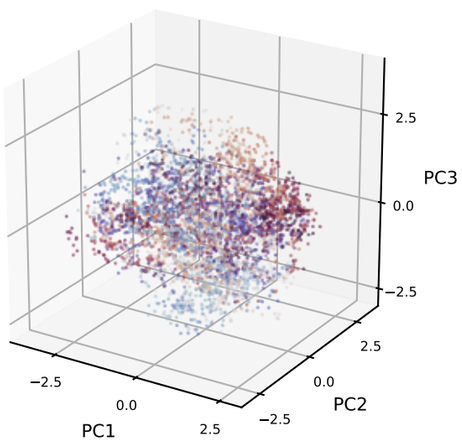


Seed 4 · PGC + grid
PC1–3: 30.9% of variance

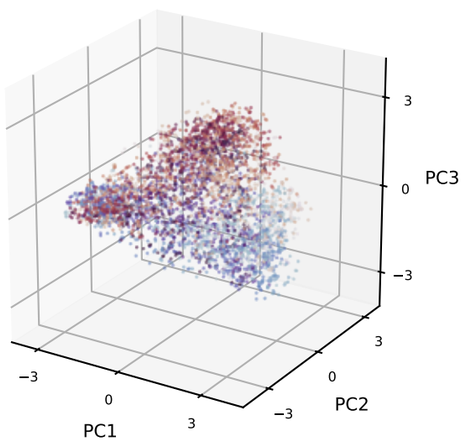




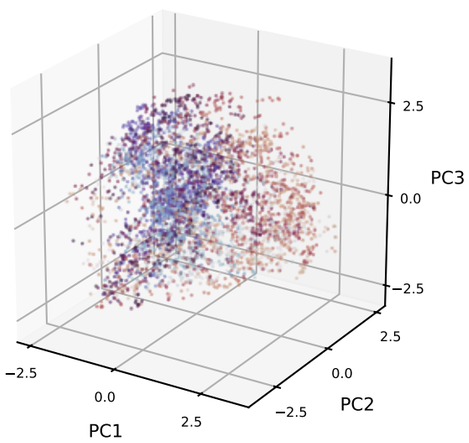
Seed 5 · Grid only
PC1–3: 23.6% of variance



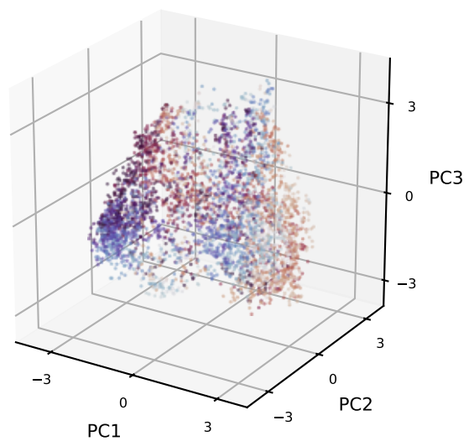
Seed 5 · PGC + grid
PC1–3: 30.3% of variance



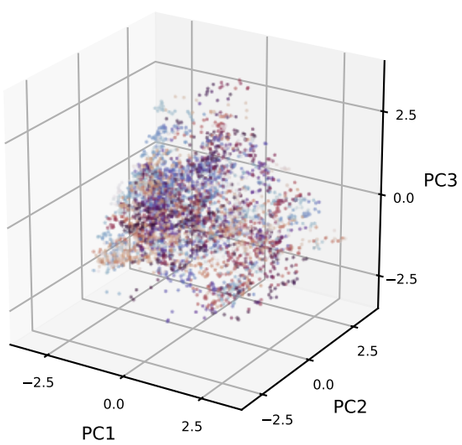
Seed 6 · Grid only
PC1–3: 24.5% of variance



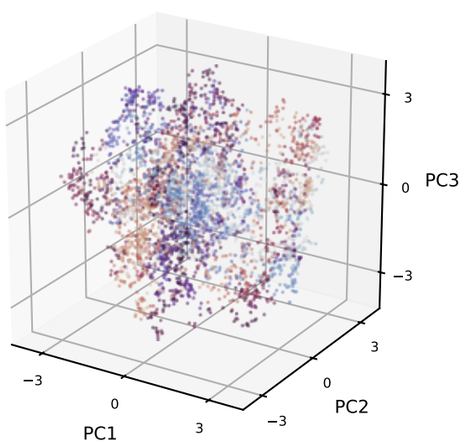
Seed 6 · PGC + grid
PC1–3: 33.1% of variance



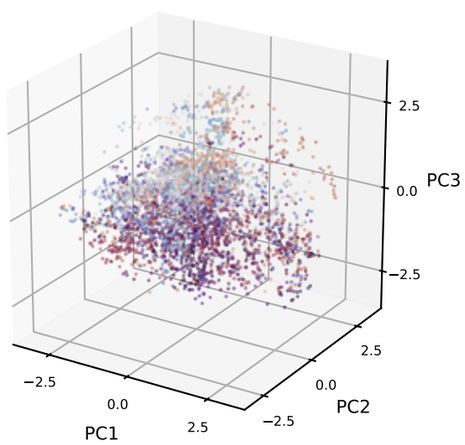
Seed 7 · Grid only
PC1–3: 21.2% of variance



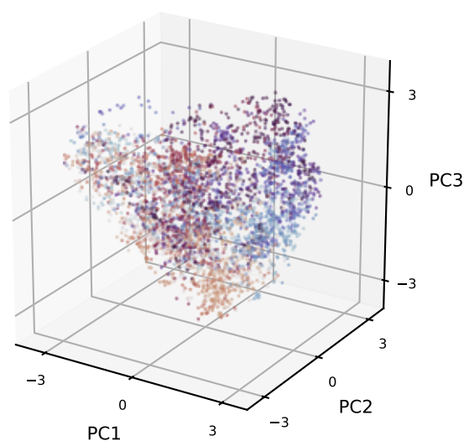
Seed 7 · PGC + grid
PC1–3: 28.3% of variance



Seed 8 · Grid only
PC1–3: 24.2% of variance



Seed 8 · PGC + grid
PC1–3: 29.5% of variance



Independent centered PCA per model; no whitening or UMAP. Equal scale within each 3-D panel; axes are not aligned across models. Shape alone is not a gridness or torus test.